

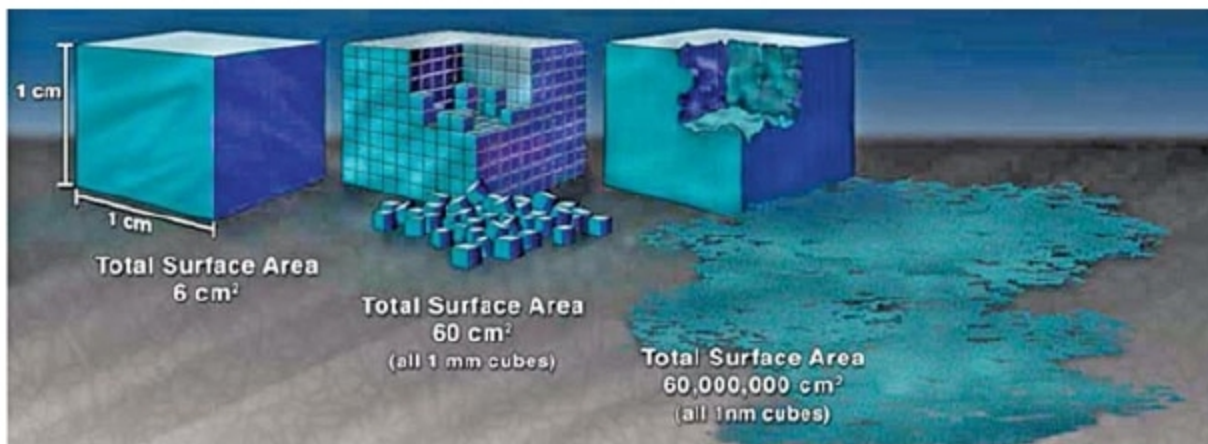


Illustration demonstrating the effect of increased surface area provided by nanostructured materials (Credit: www.nano.gov)

superior supercapacitors for energy storage.

2D materials are made up of layers of crystal, for example, molybdenum oxides. These have unique properties that encourage free flow of electrons at ultra-high speeds. Nanotubes and nanowires are 2D.

Fullerenes, dendrimers and quantum dots are 3D nanoparticles.

Carbon nanotubes (CNTs) are mechanically extremely strong, flexible and can conduct electricity. These have a range of potential applications in reinforced composites, sensors, nanoelectronics and display devices.

Inorganic nanotubes and inorganic fullerene-like materials based on layered compounds such as molybdenum disulphide have excellent lubricating properties, resistance to shockwave impact, catalytic reactivity, and high capacity for hydrogen and lithium storage.

Oxide-based nanotubes such as titanium dioxide are used for catalysis, photo-catalysis and energy storage.

Nanowires have potential applications in high-density data storage, either as magnetic read heads or as patterned storage media, and electronic and opto-electronic nanodevices for metallic interconnects of quantum devices and nanodevices.

Areas benefitting through nanotechnology

Nanotechnology is helping improve and revolutionise many technologies and industry sectors such as information technology, medicine, transportation, energy, food safety, electronics and environmental science, among others. Many everyday commercial products currently in the market and in daily use rely on nanoscale materials and processes.

Everyday materials and processes. Nanomaterials used for surface treatment helps in designing nanoscale films for eyeglasses, computer and camera displays, windows and other surfaces. These can make them water- and residue-repellent, anti-reflective, self-cleaning, resistant to ultraviolet (UV) or infrared (IR) light, anti-microbial, scratch-resistant and electrically-conductive. Nanostructured ceramic coatings have much greater toughness than conventional wear-resistant coatings.

Nanomaterials make superior household products such as stain removers, environmental sensors, air purifiers and filters, anti-bacterial cleansers, sealing products, self-cleaning house paints that resist dirt and marks, and so on. Their use in daily materials such as baseball bats, tennis rackets and helmets make these lightweight, stiff, durable and resilient.

Applications in automobiles. Lightweight structures for vehicles, aeroplanes, boats and spacecraft could lead to significant fuel savings. CNTs are a good combination of light weight and conductivity, which makes these ideal for next-generation air vehicles.

Bioengineered enzymes have been developed to convert cellulose into ethanol for fuel. Nanomaterials for automotive products are useful in high-power rechargeable battery systems and tyres with lower rolling resistance.

Nanotechnology-enabled lubricants and engine oils significantly reduce wear and tear. This has significantly extended the lifetimes of moving parts in everything from power tools to industrial machinery.

Electronics and IT applications. Nanotechnology has greatly con-

tributed to major advances in computing and electronics, leading to faster, smaller and more portable systems that can manage and store larger amounts of information. These continuously-evolving applications include transistors and basic switches that have enabled modern computing to become smaller through nanotechnology.

Entire memory can be stored on a single tiny chip using these smaller and faster transistors. Ultra-high-definition displays and televisions utilise quantum dots to produce more vibrant colours, which are more energy-efficient.

Small, flat, flexible, lightweight, non-brittle and highly-efficient electronics open the door to countless smart products.

Small sensors using silver nanowires are useful in robotics and prosthetics. Nanowire sensors can measure strain and pressure, and are flexible enough to work over a wide range of motion required by prosthetics.

Nanomaterials are used to design washable, durable and smart fabrics equipped with sensors and electronics, with capabilities for health or activity monitoring, solar energy capture and energy harvesting through movement.

Other computing and electronic products include ultra-responsive hearing aids, conductive inks for printed electronics for RFID/smart cards/smart packaging and flexible displays for e-book readers.

Medical and health applications. Better imaging and diagnostic tools enabled by nanotechnology help in earlier diagnosis and achieving better therapeutic success rates. Various nanomaterials have been developed to monitor glucose levels in people with diabetes, to block malaria parasites from spreading to new red blood cells, to purify blood as an alternative to dialysis and to stimulate immune responses.

Some other uses of nanoparticles in biology and medicine are gene delivery systems in gene therapy and biological detection of diseases. One advantage of using nanoparticles as a drug delivery system is that, targeted drugs may be developed to increase